

PLAYFAIR CIPHER

Done by

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INTRODUCTION

- A **classical encryption technique** that encrypts **pairs of letters (digraphs)** instead of single letters.
- Invented by **Charles Wheatstone (1854)** and popularized by **Lord Playfair**.
- Uses a **5 × 5 key matrix** generated from a **keyword**.
- The letters **I** and **J** are combined into a single cell.
- Provides better security than simple substitution ciphers by encrypting letter pairs.
- It belongs to the **substitution cipher family**, specifically **Polyalphabetic substitution cipher** because it encrypts **pairs of letters** rather than single letters.

PROCEDURE

- Choose a Key word (For Example MONARCHY)
- Construct a **5 × 5 Key Matrix** using the keyword and the remaining letters (combine I/J).
- Prepare the Plain text.
- Split into letter pairs.
- Check if there are any repeated letters or a single letter in the end.
- In case of repeated letters or a single letter in the end add X.

Apply Encryption Rules: Same Row → Replace each letter with the one to its **right**.

Same Column → Replace each letter with the one **below**.

Rectangle Rule → Replace each letter with the letter in the **same row** and the **other letter's column**.

EXAMPLE 1

- Here we have considered Plain text as EXPRESSIVE and Key word as MONARCHY
- Construct a 5x5 Matrix.
- Mark the keywords in the first cells of the matrix.
- Add the remaining letters in the remaining cells
- Now split the Plain Text into pairs.
- Here EX|PR|ES|SI|VE.
- Now find the pairs in the matrix and find their cipher text.

PLAIN TEXT	KEYWORD
EXPRESSIVE	MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

5x5 Matrix for keyword MONARCHY

Pair	Position	Rule Used	Cipher Pair
EX	E(3,1), X(5,4)	Rectangle	IU
PR	P(4,2), R(1,5)	Rectangle	TO
ES	E(3,1), S(4,4)	Rectangle	IL
SI	S(4,4), I(3,4)	Same Column	XA
VE	V(5,2), E(3,1)	Rectangle	UF

Encrypting each pair of the word

Plaintext	EX	PR	ES	SI	VE
Ciphertext	IU	TO	IL	XA	UF

Final Cipher Text

Example 2

- *In case of repeating letters*
- *For example,*

KeyWord: MONARCHY

Plain text: BALLET

- *After splitting the pairs, it is necessary to check*

No letter should be repeated

Letters should always be in a pair

- *To ensure the above criteria, we should add X near repeating letters or to the single letter in the end*
- *Therefore the word ballet can be written as BA/LX/LE/TX.*

KEYWORD	PLAIN TEXT
MONARCHY	BALLET

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

5x5 Matrix for keyword MONARCHY

Pair	Rule	Cipher Pair
BA	Same Column	IB
LX	Rectangle	SU
LE	Same Column	UL
TX	Same Row	LS

Encrypting each pair of the word

Plaintext	BA	LX	LE	TX
Ciphertext	IB	SU	UL	LS

Final cipher text

REAL LIFE APPLICATIONS

Military Communication – Used to encrypt confidential messages during wars.

Diplomatic Communication – Protected sensitive government communications.

Educational Purposes – Helps students learn classical cryptography and encryption techniques.

Historical Research – Studied to understand the evolution of encryption methods.

❖ **NOTE** : *The Playfair cipher is mainly used today for educational and historical purposes, as modern encryption algorithms provide much stronger security.*

ADVANTAGES

- Encrypts **pairs of letters (digraphs)** instead of single letters.
- More secure than the simple **Caesar** and **Monoalphabetic substitution** ciphers.
- Reduces the effectiveness of **single-letter frequency analysis**.
- Can be performed **manually** without computers or electronic devices.
- Simple to learn, implement, and use.

LIMITATIONS

- Requires both sender and receiver to share the **same secret keyword**.
- Uses a **5 × 5 matrix**, so I and J are usually combined.
- Extra letters (such as **X**) must be inserted between repeated letters, which can alter the original message.
- Not suitable for securing modern digital communication due to weak security compared to modern encryption algorithms.

The image features a soft, pink watercolor wash as a background. The wash is composed of several overlapping, irregular strokes of varying shades of pink, creating a textured, painterly effect. Centered over this background is the text "Thank You" in a black, elegant cursive script. The word "Thank" is on the top line, and "You" is on the bottom line, with the two words slightly offset to the right. The overall composition is simple and visually appealing, with a gentle, romantic feel.

Thank
You